NCERT Solutions for Class 12 Chemistry

Chapter 2 – Electrochemistry

INTEXT Questions

1. How would you determine the standard electrode potential of the system $Mg^{2+}|Mg$?

Ans: A cell with $Mg/MgSO_4(1M)$ as one electrode and standard hydrogen electrode $Pt, H, (1\,atm)H^+(1M)$ as the second electrode will be set up, and the emf of the cell will be measured along with the direction of deflection in the voltmeter. The direction of deflection indicates that e^- move from the magnesium electrode to the hydrogen electrode, implying that oxidation occurs on the magnesium electrode and reduction occurs on the hydrogen electrode. As a result, the cell can be represented in the following way:

$$Mg\left|Mg^{2+}(1M)\right\| H^+(1M)\left|H^2,(1atm)Pt\right.$$

$$E^o_{cell} = E^o_{H+\frac{1}{2}} - E^o Mg^{2+}/Mg$$

$$\text{put } E^o_{H+\frac{1}{2}H_2} = 0$$

$$\therefore E^o_{Mg^{2+}/Mg} = -E^o{}_{cell}$$

2. Can you store copper sulphate solutions in a zinc pot?

Ans: Cu is less reactive than Zn, so Cu is easily shifted from $CuSO_4$ solution in the following reaction:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO4_{(qq}\big) + Cu(s)$$

If we express it in relations of emf, it will be like:

$$Ecell^{\circ} = Ecu^{2+}/Cu^{\circ} - E^{\circ}zn^{2+}/Zn$$

$= 0.34\ V - (-0.76\ V) = 1.10\ V$

As, The positive E_{cell}° value indicates the occurrence of spontaneous reactions. if $CuSO_4$ is stored in Zn pot, Zn will react will Cu and originality of $CuSO_4$ will not be maintained.

3. Consult the table of standard electrode potentials and suggest three substances that can oxidize ferrous ions under suitable conditions.

Ans: In the process of oxidation of Fe^{2+} to Fe^{3+}, i.e., $Fe^{2+} \rightarrow Fe^{3+} + e^{-}$; the reduction potential value is negative in nature i.e. $E^{o}ox = -0.77\ V$. Only compounds with powerful oxidizing agents and positive reduction potentials larger than $0.77\ V$ may oxidize Fe^{2+} to Fe^{3+}, resulting in a positive emf of the cell reaction. This is true for elements like halogens Br_2, Cl_2 and F_2 in the series below Fe^{3+}/Fe^{2+}.

4. Calculate the potential of hydrogen electrode in contact with a solution whose pH is 10 .

Ans: For hydrogen electrode, $H^{+} + e^{-} \rightarrow 1/2H_2$

Given: $pH = 10$

Using Nernst equation,

$$E_{H^{-}\frac{1}{2}H_2} = E^{o}_{H^{+}\frac{1}{2}H_2} - \frac{0.0591}{n}\log\frac{1}{\left[H^{+}\right]}$$

$$0 - \frac{0.0591}{1}\log\frac{1}{10^{-10}}$$

$$\left\{\begin{array}{c} pH = 10 \\ \left[H^{+}\right] = 10^{-10}M \end{array}\right\} = -0.519 \times 10$$

$= 0.591\ V$

Hence, the potential of hydrogen electrode in contact with a solution whose pH is 10 is $0.591\ V$.

5 Calculate the emf of the cell in which the following reaction takes place: $Ni_{(a)} + 2Ag^{+}(0.002M) \rightarrow Ni^{i+}(0.160M) + 2Ag_{(a)}$ **Given that** $= E^{\circ}(coll) 1.05\ V$**.**

Ans: Using Nernst equation,

$$E_{cell} = E^{o}_{cell} - \frac{0.0591}{n} \log \frac{\left[Ni^{2+}\right]}{\left[Ag^{+}\right]^{2}}$$

$$= 1.05\ V - \frac{0.0519}{2} \log \frac{0.160}{(0.002)^{2}}$$

$$= 1.05 - \frac{0.0591}{2} \log\left(\log 4 \times 10^{4}\right)$$

$$= 1.05 - \frac{0.0591}{2}(4.6021)$$

$$= 1.05 - 0.14\ V = 0.91\ V$$

Therefore, the emf of the cell is $0.91\ V$.

6. $0.236\ V$ **at** $298\ K$**. Calculate the standard Gibbs energy and the equilibrium constant of the cell reaction.**

Ans: $2Fe^{3} + (zg) + 2I^{-}(zq) \rightarrow 2Fe^{2} + (zg) + I_{2}(g)$

For the given cell, the number of transacted electrons i.e. $n = 2$

Using the given formula;

$$\Delta_r G^{o} = -FE^{o}{}_{cell}$$

$$= -2 \times 96500 \times 0.236$$
$$= -45.55\ kJ\ mol - 1$$

Also, $\Delta_r G^{\circ} = -2.303 RT \log KC$

$$\Rightarrow \log K_c = \frac{\Delta, G^{o}}{2.303RT} = \frac{-45.55}{2.303 \times 8.314 \times 10^{-3} \times 298} = 7.983$$

$\Rightarrow K_c = \text{antilog}(7.983)$

$= 9.616 \times 10^7$

7. Why does the conductivity of solution decrease with dilution?

Ans: The conductance of ions contained in a unit volume of solution is called conductivity. The number of ions per unit volume decreases with increase in dilution. As a result, the conductivity drops.

8. Suggest a way to determine the value Λ° of water.

Ans: By using Kohlrausch's law for H_2O, we can calculate Λm°

The Kohlrausch Law says that when dissociation is complete at infinite dilution, each ion adds a definite amount to the electrolyte's equivalent conductance, regardless of the nature of the ion with which it is involved, and the value of equivalent conductance at infinite dilution for any electrolyte is the sum of contributions of its ionic species (cations and anions).

$$\Lambda m^\circ = \Lambda m^\circ(\text{Hcl}) + \Lambda m^\circ(\text{NaOH}) - \Lambda m^\circ(\text{NaCl})$$

The Λ° values of HCl, NaOH and NaCl are known as they are strong electrolytes and dissociates completely.. By putting their values in the above equation, we can have value of Λm° for H_2O.

9. The molar conductivity of $0.025\,\text{mol}\,\text{L}^{-1}$ methanoic acid is $46.1\,\text{S}\,\text{cm}^2\,\text{mol}^{-1}$. Calculate its degree of dissociation and dissociation constant Given $\lambda^0\left(H^+\right) = 349.6\,\text{cm}^2$ and $\lambda^\circ\left(\text{HCOO}^-\right) = 54.6\,\text{cm}^2\,\text{mol}^{-1}$

Ans:

$\Lambda_m^o(HCOOH)\lambda^\circ\left(H^+\right) + \lambda^\circ(HCOO)^-$

$= 349.6 + 54.6$

$= 404.2\,\text{S}\,\text{cm}^2\,\text{mol}^{-1}$

$\Lambda_m^c = 46.1 S\,cm^2\,mol^{-1}$

$\therefore \alpha \therefore \frac{\Lambda_m^c}{\Lambda_m^c} = \frac{46.1}{404.2} = 0.114$

$HCOOH \rightleftharpoons HCOO^- + H^+$

Initial conc	c	0	0
At equi,	$c(1-\propto)$	$c \propto$	$c \propto$

$\therefore K_a = \frac{c\alpha \cdot c\alpha}{c(1-\alpha)} = \frac{c\alpha^2}{1-\alpha}$

$= \frac{0.025 \times (0.114)^2}{1-0.114} = 3.67 \times 10^{-4}$

10. If a current of 0.5 ampere flows through a metallic wire for 2 hours, then how many electrons would flow through the wire?

Ans: As from the formula,

$Q = It$

$= 0.5 \times (2 \times 60 \times 60) = 3600C$

$1F \Rightarrow 96500C \Rightarrow 1 mole\ of\ e^{-1}s$ $\therefore 3600C$ is equivalent to the flow of $e^{-1}s$

$\therefore 6.02 \times 10^{23} e^{-1}s$

$= \frac{6.02 \times 10^{23}}{96500} \times 3600 = 2.246 \times 10^{22} e^{-1} s$

11. Suggest a list of metals that are extracted electrolytically.

Ans: Na, Ca, Mg, and Al

12. Consider the reaction: $CrO_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$**. What is the quantity of electricity in coulombs needed to reduce** $1\,mol$ **of** $Cr_2O_7^{2-}$

Ans: For the above reaction to take place, $1\,mol$ of $Cr_2O_7^{2-}$ will require $6\,F$ $= 6 \times 96500 = 579000C$ 0f electricity.

Hence, $579000C$ of electricity are required for reduction of $Cr_2O_7^{2-}$ to Cr^{3+}

13. Write the chemistry of recharging the lead storage battery, highlighting all the materials that are involved during recharging.

Ans: A lead storage battery has a lead anode, a lead cathode with lead dioxide packed (PbO_2) in it, and a $38\% H_2SO_4$ solution as the electrolyte. When the battery is in operation, the following reactions occur:

At Anode: $Pb_{(s)} + SO_4^{2-} \rightarrow PbSO_{4(s)} + 2e^-$

At cathode: $PbO_{2(s)} + SO_4 2\text{-}(ag) + 4H^+{}_{(aq)} + 2e^- \longrightarrow PbSO_{4(s)} + 2H_2O_{(l)}$

Complete reaction: $Pb_{(s)} + PbO_{2(s)} + 2H_2SO_{4(aq)} \rightarrow 2PbSO_{4(s)} + 2H_2O_{(l)}$

The reverse process occurs when the battery is charged, in which $PbSO_4$ deposited on the electrodes is transformed back to Pb and PbO_2, and H_2SO_4 is restored.

14. Suggest two materials other than hydrogen that can be used as fuels in fuel cells.

Ans: Methane and Methanol.

15. Explain how rusting of iron is envisaged as setting up of an electro chemical cell.

Ans: The Water present on the surface of iron dissolves acidic oxides of air like CO_2, SO_2 etc, to form acids which dissociate to give H^+ ions:

$$H_2O + CO_2 \rightarrow H_2CO_3 \rightleftharpoons 2H^+ + CO_3^{2-}$$

In the presence of H^+, iron Fe^{2+} releases e^{-1} to form Fe^{3+}.

Hence, this acts as anode:

$$Fe_{(s)} \rightarrow Fe^{2+}{}_{(aq)} + 2e^-$$

The e^{-1} lost then travel through the metal to reach place where these electrons are utilized by H^+ ions and dissolved oxygen and reduction takes place. Thus acts as cathode:

$$O_2(g) + 4H^+_{(aq)} + 4e^- \rightarrow 2H_2O_{(g)}$$

The overall reaction is:

$$2Fe_{(s)} + O_{2(g)} + 4H^+_{(aq)} \rightarrow 2Fe^{2+}_{(aq)} + 2H_2O_{(l)}$$

As a result, an electrochemical cell is established on the surface. The atmospheric oxygen oxidizes ferrous ions to ferric ions, which mix with water to produce hydrated ferric oxide., $Fe_2O_3.XH_2O$ which is known as rust.

NCERT Exercise

1. Arrange the following metals in the order in which they displace each other from the solution of their salts: Al, Cu, Fe, Mg **and** Zn

Ans: Mg, Al, Zn, Fe, Cu, Ag

2. Given the standard electrode potentials, $K^+ / K = 2.93\ V, Ag^+ / Ag = 0.80\ V$

$Hg^{2+} / Hg = 0.79\ V$

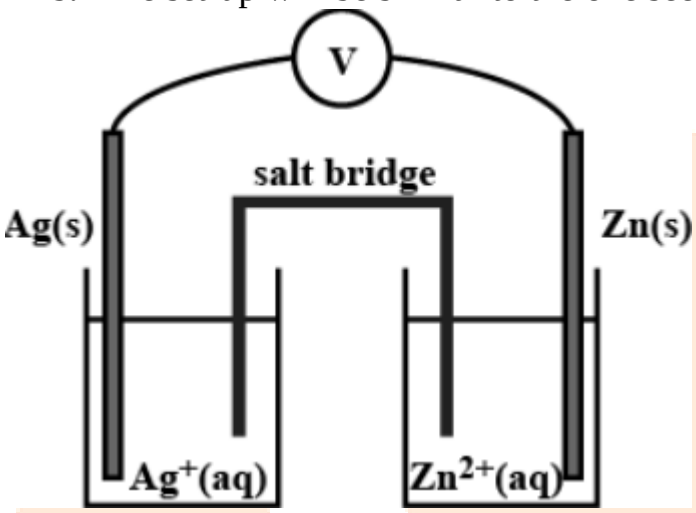
$Mg^{2+} / Mg = -2.73\ V, Cl^{3+} / Cr = 0.74\ V$

Arrange these metals in their increasing order of reducing power.

Ans: The larger the oxidation potential, the easier it is to oxidize it, and hence the higher the reducing power. As a result, the reducing power of the elements will be in the following ascending sequence: $Ag < Hg < Cl < Mg < K$.

3. Depict the galvanic cell in which the reaction, $2Ag^+(aq) \longrightarrow Zn^{2+}(aq) + 2Ag(s)$ **takes place. Further show:**

Ans: The set-up will be similar to the one seen below.



(i) Which of the electrode is negatively charged?

Ans: The anode, or zinc electrode, will have a negative charge.

(ii) The carriers of the current in the cell.

Ans: In the external circuit, current will travel from silver to copper.

(iii) Individual reaction at each electrode.

Ans: At anode: $Zn\left((n) \rightarrow Zn^{2+}(q) + 2e^{-}\right.$

At cathode: $2Ag^{+}{}_{(zq)} + 2e^{-} \rightarrow 2Ag_{(i)}$

4. Calculate the standard cell potentials of galvanic cell in which the following reactions take place

(i) $2Cr(s) + Cd^{2+}_{(aq)} \rightarrow 2Cr^{3+}{}_{(aq)} + 3Cd_{(s)}$

(ii) $Fe^{2+}_{(aq)} + Ag^{+}_{(aq)} \rightarrow Fe^{3+}_{(aq)} + Ag(s)$

Calculate the $\Delta_r G^o$ **and equilibrium constant of the reactions**

Given

$E^o_{Cr^3,Cr} = -0.74\,V$

$E^o_{Cr^{2+},Cd} = -0.40\,V$

$E^o_{Ag^+}Ag = 0.80\,V$

$E^o_{Fe^3,Fe^{2+}} = 0.77\,V$

Ans:

E° cell $= E^\circ$ cathode $- E^\circ$ Anode

$= -0.40\,V - (-0.74\,V) = +0.34\,V$

$\Delta_r G^o = nFE^\circ$ cell

$= -6 \times 96500 Cmol^{-1} \times 0.34\,V$

$= 196860 CVmol^{-1}$

$= 196860\,J\,mol^{-1}$

$= -196.86\,kJ\,mol^{-1}$

$-\Delta_r G^\circ = 2.303x8.314 \times 298 \log K$

$196860 = 2.303 \times 8.314 \times 298 \log K$

or $\log K = 34.5014$

$K =$ Antilog $34.5014 = 3.172 \times 10^{34}$

$E^\circ cell = +0.80\,V - 0.77\,V = +0.03\,V$

$\Delta_r G^o = nFE^\circ$ cell

$= -1 \times \left(96500 CVmol^{-1}\right) \times (0.03\,V)$

$= -2.895 CVmol^{-1} = -2895\,J\,mol^{-1}$

$= 2.895\,kJ\,mol^{-1}$

$\Delta_r G^o = 2.303RT\log K$

$-2895 = -2.303 \times 8.314 \times 298 \times \log K$

or $\log K = 0.5074$

or $K = \text{Antilog}(0.5074) = 3.22$

5. Write the Nernst equation and emf of the following cells at $298\,K$.

(i) $Mg_{(s)} | Mg^{2+}(0.001M) || Cu^{2+}(0.0001M)Cu_{(s)}$

Ans: Cell reaction:

$Mg + Cu^{2+} \rightarrow Mg^{2+} + Cu(n = 2)$

Applying Nernst equation:

$$E_{cell} = ECell° - \frac{0.0591}{2}\log\frac{[Mg^{2+}]}{[Cu^{2+}]}$$

$$E_{cell} = 0.34 - (-2.37) - \frac{0.0591}{2}\log\frac{10^{-3}}{10^{-4}}$$

$= 2.71 - 0.02955 = 2.68\ V$

(ii) $Fe_{(s)} | Fe^{2+}(0.001M) || H^{+}(1M)H_2(g)(1bar)Pt_{(S)}$

Ans: Cell reaction:

$Fe + 2H^{+} \rightarrow Fe^{2+} + H_2(n = 2)$

Applying Nernst equation

$$E_{cell} = E^o_{cell} - \frac{0.0591}{2}\log\frac{[Fe^{2+}]}{[H^{+}]^2}$$

$$E_{cell} = 0 - (-0.44) - \frac{0.0591}{2}\log\frac{10^{-3}}{(1)^2}$$

$$= 0.44 - \frac{0.0591}{2} \times (-3)$$

$$= 0.44 + 0.0887 = 0.5287 \text{ V}$$

(iii) $Sn_{(s)} \left| Sn^{2+}(0.050M) \right\| H^{+}(0.020M) \left| H_2(g)(1bar)Pt_{(S)} \right.$

Ans: Cell reaction:

$$Sn + 2H^{+} \rightarrow Sn^{2+} + H_2 (n = 2)$$

Applying Nernst equation:

$$E_{cell} = Ecell° - \frac{0.0591}{2} \log \frac{\left[sn^{2+}\right]}{\left[H^{+}\right]^2}$$

$$E_{cell} = Ecell° - \frac{0.0591}{2} \log \frac{0.05}{(0.02)^2}$$

$$= 0 - (-0.14) - \frac{0.0591}{2} \log \frac{0.05}{(0.02)^2}$$

$$= 0.14 - \frac{0.0591}{2} \log 125$$

$$= 0.14 - \frac{0.0591}{2} (2.0969) = 0.078 \text{ V}$$

(iv) $Pt_{(s)} \left| Br_{2(I)} \right| Br - (0.010M) \| H + (0.030M) H_{2(g)} (1bar) Pt_{(S)}$

Ans: Cell reaction:

$$2Br + 2H^{+} \rightarrow Br_2 + H_2 (n = 2)$$

Applying Nernst equation:

$$E_{cell} = Ecell° - \frac{0.0591}{2} \log \frac{1}{\left[Br^{-}\right]^2 \left[H^{+}\right]^2}$$

$$= (0 - 1.08) - \frac{0.0591}{2} \log \frac{1}{(0.01)^2 (0.03)^2}$$

$$= -1.08 - \frac{0.0591}{2}\log\left(1.111\times10^{7}\right)$$

$$= -1.08 - \frac{0.0591}{2}(7.0457)$$

$$= -1.08 - 0.208 = -1.288\ \text{V}$$

As a result, oxidation will take place at the hydrogen electrode, whereas reduction will take place at the Br_2 electrode.

6. In the button cells widely used in watches and other devices the following reaction takes places:

Determine ΔrG° and E° for the reaction.

Given

$$Zn \rightarrow Zn^{2+} + 2e^{-}, E^{\circ} = 0.76\ \text{V}$$

$$Ag_2O + H_2O + 2e^{-} \rightarrow 2Ag + 2OH^{-}, E^{\circ} = 0.344\text{v}$$

Ans: In the given reaction, Zn is getting oxidized and Ag_2O is being reduced.

$$E^{\circ}\text{cell} = 0.344 + 0.76 = 1.104\ \text{V}$$

$$\Delta G^{\circ} = nFE^{\circ}\text{cell} = -2\times96500\times1.104\ \text{J}$$

$$\Delta G^{\circ} = -2.13\times10^{5}\ \text{J}$$

7. Define conductivity and molar conductivity for the solution of an electrolyte. Discuss their variation with concentration.

Ans: The conductivity of a solution is defined as the conductance of a solution with a length of $1\,\text{cm}$ and a cross-sectional area of $1\ \text{s}\,\text{cm}$.

When the electrodes are one cm apart and the area of cross-section of the electrodes is large enough that the entire solution is contained between them, the molar conductivity of a solution at a dilution (V) is the conductance of all the ions produced

from one mole of the electrolyte dissolved in the solution Vcm^3. Λ_m is the most common symbol for conductivity.

The conductivity of a solution (for both strong and weak electrolytes) diminishes as the concentration of the electrolyte drops, i.e. as dilution occurs. This is owing to the fact that when the solution is diluted, the number of ions per unit volume of the solution decreases.

On dilution, the molar conductivity of a solution increases as with decrease in concentration of the electrolyte. This is owing to the fact that when the solution is diluted, the number of ions per unit volume of the solution decreases. The molar conductivity of a solution increases as the electrolyte content drops. This is because dilution increases both the quantity of ions and their mobility. The molar conductivity is known as limiting molar conductivity as the concentration approaches zero.

8. The conductivity of $0.20M$ solution of KCl at $298\,K$ is $0.0248\,S\,cm^{-1}$. Calculate its molar conductivity.

Ans: $\Lambda_m = \dfrac{k \times 1000}{\text{Molanity}} = \dfrac{0.2485\,cm^{-1} \times 1000\,cm^3 L^{-1}}{0.20 moL^{-1}} = 124\,Scm^2 mol^{-1}$

9. The resistance of a conductivity cell containing $0.001MKCl$ solution at $298\,K$ is 1500Ω. What is the cell constant if conductivity of $0.001MKCl$ solution at $298\,K$ is $0.146 \times 10^{-3}\,S\,cm^{-1}$?

Ans: From the following relation:

$$\text{Cell constant} = \frac{\text{Conductivity}}{\text{Conductance}} = \text{Conductivity} \times \text{Resistance}$$

$$= 0.146 \times 10^{-3}\,S\,cm^{-1} \times 1500\Omega$$

$$= 0.218\,cm^{-1}$$

10. The conductivity of $NaCl$ at $298\,K$ has been determined at different concentrations and the results are given below:

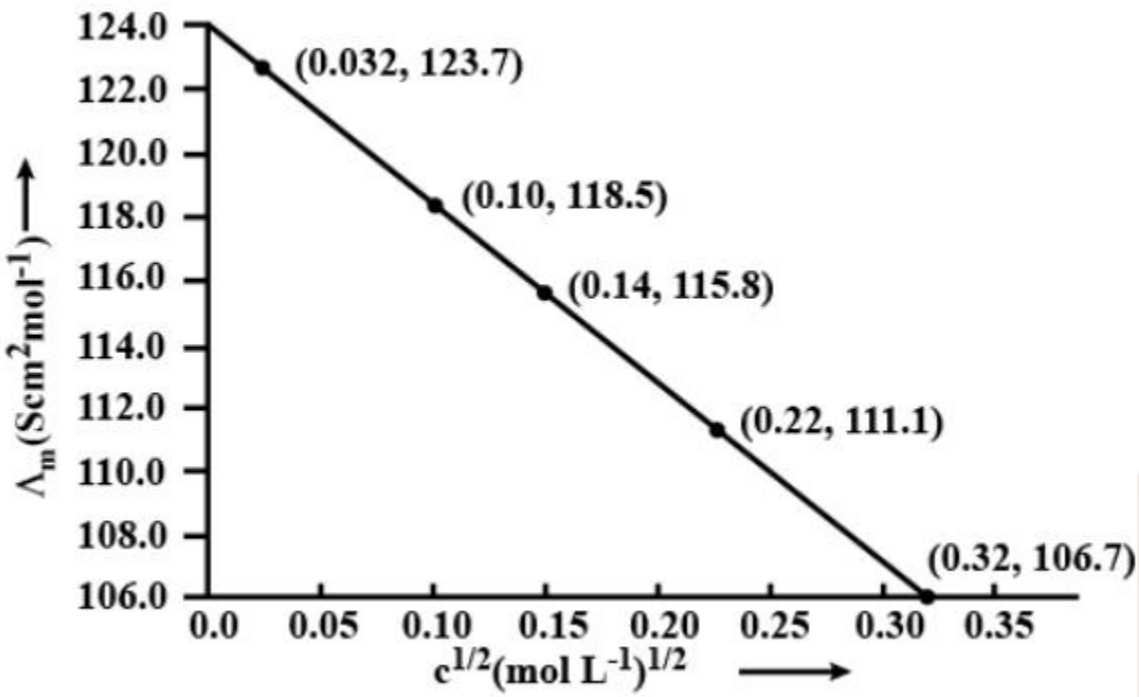
concentration/M	0.001	0.010	0.020	0.050	0.100
$10^2 \times$ K/S m^{-1}	1.237	11.85	23.15	55.53	106.74

For all concentrations and draw a plot between Λ_m and $C^{1/2}$

Find the value of Λ_m°

Ans: Using the following unit conversion factor, $\frac{1S\ cm^{-1}}{100Sm^{-1}} = 1$

Concentration (M)	$K\left(Sm^{-1}\right)$	$K\left(Scm^{-1}\right)$	$\Lambda_m = \frac{1000 \times k}{\text{Molanity}}\left(Scm^2\ mol^{-1}\right)$	$C^{1/2}\left(M^{1/2}\right)$
10^{-3}	1.237×10^{-2}	1.237×10^{-4}	$\frac{1000 \times 1.237 \times 10^{-4}}{10^{-3}} = 123.7$	0.0316
10^{-2}	11.85×10^{-2}	11.85×10^{-4}	$\frac{1000 \times 11.85 \times 10^{-4}}{10^{-2}} = 1118.5$	0.100
2×10^{-2}	23.15×10^{-2}	23.15×10^{-4}	$\frac{1000 \times 23.15 \times 10^{-4}}{2 \times 10^{-2}} = 115.8$	0.141
5×10^{-2}	55.53×10^{-2}	55.53×10^{-4}	$\frac{1000 \times 55.53 \times 10^{-4}}{5 \times 10^{-2}} = 111.1$	0.224
10^{-1}	106.74×10^{-2}	106.74×10^{-4}	$\frac{1000 \times 106.74 \times 10^{-4}}{10^{-1}} = 106.7$	0.316



Λ°cm = Intercept of Λm axis $= 124.0\ \mathrm{S\ cm^2\ mol^{-1}}$, Extrapolation to zero concentration yields this result.

11. Conductivity of 0.00241M **acetic acid is** $7.896\times10^{-5}\mathrm{Scm^{-1}}$. **Calculate its. Calculate its molar conductivity. If** Λ°m **for acetic acid is** $390.5\ \mathrm{S\ cm^2\ mol^{-1}}$**, what is its dissociation constant?**

Ans: $\Lambda_m^c = \dfrac{\kappa \times 1000}{\text{Molarity}}$

$$= \frac{\left(7.896\times10^{-5}\ \mathrm{S\ cm^{-1}}\right)\times1000\ \mathrm{cm^3L^{-1}}}{0.241\ \mathrm{molL^{-1}}} = 32.76\ \mathrm{S\ cm^2\ mol^{-1}}$$

$$\alpha = \frac{\Lambda_m^c}{\Lambda m^{\circ}} = \frac{32.76}{390.5} = 8.4\times10^{-2}$$

$$k_a = \frac{C\alpha^2}{1-\alpha} = \frac{0.24\times\left(8.4\times10^{-2}\right)^2}{1-0.084} = 1.86\times10^{-5}$$

12. How much charge is required for the following reductions:

(i) $1\,\text{mol of } Al^{3+} \text{ to Al}$ **?**

Ans: Aluminum ion have to lose three electrons to be in aluminum elemental state and the reaction is $Al^{3+} + 3e \rightarrow Al$. The amount of charge needed for reduction of $1\,\text{mol}$ will be: $Al^{3+} = 3\,F = 3 \times 96500C = 289500C$.

(ii) $1\,\text{mol of } Cu^{2+} \text{ to Cu}$ **?**

Ans: cupric ion needs to lose two electrons to be in copper (0) state. The electrode reaction occurring is as follows: $Cu^{2+} + 2e^{-} \rightarrow Cu$

The amount of charge required for conversion of
$1\,\text{mol of } Cu^{2+} = 2\,F = 2 \times 96500 = 193000C$.

(iii) $1\,\text{mol of } MnO^{4} \text{ to } Mn^{2+}$

Ans: in the compound MnO^{4}, manganese is in +7 oxidation state; which is reduced to +2 oxidation state with the release of five electrons.

$MnO_4 \rightarrow Mn^{2+}$ i.e. $Mn^{7+} + 5e^{-} \rightarrow Mn^{2+}$

The Quantity of charge needed for reduction
of $1\,\text{mol of } MnO^{4} \text{ to } Mn^{2+} = 5\,F = 5 \times 96500C = 4825000C$.

13. How much electricity in terms of Faraday is required to produce.

(i) $20.0\,\text{g of Caa from molten } CaCl_2$ **?**

Ans: $Ca^{2+} + 2e^{-} \rightarrow Ca$

As There are two electrons transacted in the above reaction, therefore $1\,\text{mol}$ of Ca or $40\,g$ of Ca will require $= 2\,F$ electricity and $20\,g$ of Ca will require $= 1\,F$ of electricity.

(ii) $40.0\,\text{g of Al from molten } Al_2O_3$ **?**

Ans: $Al^{3+} + 3e^{-} \rightarrow Al$

As the number of electrons transacted in the above reduction of aluminum ion is three,

Therefore, $1 \text{ mol of Al or } 27 \text{ g of Al}$ will require $= 3\,\text{F}$ electricity

And, $40\,\text{g}$ of Al will require electricity

$= \frac{3}{27} \times 40 = 4.44\text{F}$ of electricity.

14. How much electricity is required in coulomb for the oxidation of

(i) $1\,\text{mol of } H_2O \text{ to } O_2$

Ans: The oxidation of water occurs in the following way:

$$H_2O \longrightarrow H_2 + \frac{1}{2}O_2$$

Or $O^{2-} \longrightarrow \frac{1}{2}O_2 + 2e^-$

The number of electrons transacting are two. Therefore the amount of electricity needed $= 2\,\text{F} = 2 \times 96500\text{C} = 193000\text{C}$

(ii) $1\,\text{mol of FeO to } Fe_2O_3$

Ans: The oxidation reaction of FeO takes place in the following manner:

$$FeO + \frac{1}{2}O_2 \rightarrow \frac{1}{2}Fe_2O_3$$

Or $Fe^{2+} \rightarrow Fe^{3+} + e^-$

The electron transfer is of one electron unit, therefore the Quantity of electricity essential $= 1\,\text{F} = 96500\text{C}$

15. A solution of $Ni(NO_3)_2$ is electrolyzed between platinum electrodes using a current of 5 amperes for 20 minutes. What mass of Ni is deposited at the cathode?

Ans: Firstly, we will find charge in coulombs using the formula: $Q = It$

Amount of electricity passed $(Q) = (5\ A) \times (20 \times 60 sec) = 6000C$

The reduction of nickel occurs in the following way:

$Ni^{2+} + 2e^{-} \longrightarrow Ni$

Therefore, $2\ For\ 2 \times 96500C$ is the amount of charge deposit for $1\ mol$ of $Ni = 58.7\ g$

Now,

The $6000C$ of charge will deposit $= \frac{58.7 \times 6000}{2 \times 96500} = 1.825\ g\ of\ Ni$

16. Three electrolytic cells, A, B, C containing solutions of $ZnSO_4, AgNO_3$ and $CuSO_4$, respectively are connected in series. A steady current of 1.5amperes was passed through them until $45\ g$ of silver deposited at the cathode of cell B. How long did the current flow? What mass of copper and zinc were deposited?

Ans: Given: $I = 1.5\ A, W = 1.5\ g$ of $Ag\ t = ?, E = 108, n = 1$

Applying Faraday's first law of electrolysis,

$W = ZIt$

Or

$$W = \frac{E}{nF} It$$

By substituting the values provided in the about formula, t can be calculated;

$$t = \frac{1.45 \times 96500}{1.5 \times 108} = 863.73 \text{ seconds.}$$

Now for $Cu, W_1 = 1.45\ g(given)$ of $AgE_1 = 108, W_2 = ?$

$E_2 = 31.75$

Applying Faraday's 2nd law of electrolysis

$$\frac{W_1}{W_2} = \frac{E_1}{E_2}$$

$$\frac{1.45}{W_2} = \frac{108}{31.75}$$

$$W_2 = \frac{1.45 \times 31.75}{108}$$

Copper deposited $= 0.426\,g$ of Cu

likewise, for $Zn, W_1 = 1.45\,g$ of $Ag, E_1 = 108, W2 = ?\,E2 = 32.65$

Using formula, $\frac{W_1}{W_2} = \frac{E_1}{E_2}$

$$\frac{1.45}{W_2} = \frac{108}{32.65}$$

$$W_2 = \frac{1.45 \times 32.65}{108} = 0.438 \text{ of Zn}$$

Zinc deposited by $1295.6g\ C$ is 0.438 g.

17. Predict if the reaction between the following is feasible:

Given standard electrode potentials:

$E^o_{1/2, I_2, I^-} = +0.451V$

$E^o_{Cu^{2+}, Cu} = +0.34V$

$E^o_{1/2 Br_2, Br^-} = +1.09V$

$E^o_{Ag^+, Ag} = +0.80V, E^o_{Fe^{3+}, Fe^{2+}} = +0.77V$

(i) Fe^{3+} (aq) **and** I^-(aq)

Ans: The reaction is feasible if the emf of the cell reaction is positive.

$Fe^{3+}(aq)+I^{-}_{(aq)} \rightarrow Fe^{2+}(aq)+\frac{1}{2}I^{2}(g)$

i.e., $Pt/I^{2}/I^{-}(aq) \parallel Fe^{3+}(aq)\left|Fe^{2+}(aq)\right|Pt$

$\therefore E^{o}_{cell} = E_{Fe^{3+}\cdot Fe^{2+}} - E^{o}_{1/2I_2, f^2}$

$= 0.77 - 0.54 = 0.23V$ (Feasibles)

(ii) $2Ag^{+}{}_{(aq)} + Cu_{(s)} \rightarrow 2Ag_{(s)} + Cu^{2+}_{(aq)}$

Ans: $2Ag^{+}{}_{(aq)} + Cu_{(s)} \rightarrow 2Ag_{(s)} + Cu^{2+}_{(aq)}$

i.e., $Cu\left|Cu^{2+}(aq) \parallel Ag^{+}_{(aq)}\right|Ag$

$\therefore E^{o}_{cell} = E^{o}{}_{Ag^{+}Ag} - E^{o}_{Cu}, Cu$

$= 0.80 - 0.34 = 0.46V$ (Feasible)

(iii) $Fe^{3+}(aq)+Br^{-}_{(aq)} \rightarrow Fe^{2+}{}_{(aq)} + \frac{1}{2}Br_{2(g)}$

Ans: $Fe^{3+}(aq)+Br^{-}_{(aq)} \rightarrow Fe^{2+}{}_{(aq)} + \frac{1}{2}Br_{2(g)}$

$E^{o}_{cell} = 0.77 - 1.09 = -0.32v$(Not feasible)

(iv) $Ag_{(s)} + Fe^{3+}_{(aq)} \rightarrow Ag^{+}_{(aq)} + Fe^{2+}(aq)$

Ans: $Ag_{(s)} + Fe^{3+}_{(aq)} \rightarrow Ag^{+}_{(aq)} + Fe^{2+}(aq)$

$E^{o}_{cell} = 0.77 - 0.80 = -0.03v$(not feasible)

(v) $\frac{1}{2}Br_{2(g)} + Fe^{2+}(aq) \rightarrow Br^{-}_{(aq)}Fe^{3+}(aq)$

Ans: $\frac{1}{2}Br_{2(g)} + Fe^{2+}(aq) \rightarrow Br^{-}_{(aq)}Fe^{3+}(aq)$

$E^{o}_{cell} = 1.09 - 0.77 = 0.32V$(Feasible)

18. Predict the products of electrolysis of the following:

(i) An aqueous solution of $AgNO_3$ with silver electrodes.

Ans: $AgNO_3(s) \longrightarrow Ag^+(aq) + NO_3^-(aq)$

$H_2O \leftrightarrows H^+ + OH^-$

At cathode: The discharge potential of Ag^+ ions is lower than that of H^+ ions. As a result, Ag ions will be deposited as Ag rather than H^+ ions.

At anode: When the Ag anode is attacked by NO_3^- ions, the Ag dissolves and forms ions in the solution.

$Ag(s) \longrightarrow Ag^+(aq) + e^-$

(ii) An aqueous solution of $AgNO_3$ with platinum electrodes.

Ans: At cathode: The discharge potential of Ag^+ ions is lower than that of H^+ ions. As a result, Ag ions will be deposited as Ag rather than H^+ ions.

At anode: Because the anode is not attackable, OH ions have a lower discharge potential than NO_3^- ions. As a result, OH will be evacuated first, followed by NO_3^- ions, which will disintegrate and release O_2.

$OH^- \rightarrow OH + e^-$

$4OH \rightarrow 2H_2O_{(1)} + O_{2(g)}$

(iii) A dilute solution of H_2SO_4 with platinum electrodes.

Ans: $H_2SO_4 \rightarrow 2H^+ + SO_4^{2-}$

$H_2O \rightleftharpoons H + OH^-$

At cathode: $H^+ + e^- \rightarrow H$

$H + H \rightarrow H_{2(g)}$

At anode: $OH^- \rightarrow OH + e^-$

$4OH \rightarrow 2H_2O + O_{2(g)}$

Therefore, Hydrogen gas is released at the cathode and O_2 gas at the anode.

(iv) An aqueous solution of $CuCl_2$ with platinum electrodes.

Ans: $CuCl_2 \rightarrow Cu^{2+} + 2Cl^-$

$H_2O \rightleftharpoons H^+ + OH^-$

At cathode: Cu^{2+} ions will be reduced more than H^+ ions, resulting in copper deposition at the cathode.

$Cu^{2+} + 2e^- \rightarrow Cu$

At anode: Cl^- ions will be released first, followed by OH^- ions that will stay in solution. $Cl^- \rightarrow Cl + e^-$

$Cl + Cl \rightarrow Cl_{2(g)}$

Hence, copper will get set down on the cathode and Cl_2 gas will be liberated at the anode.